

Lec 38 Conditional GAN

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Transcript

[00:00] Welcome to the lecture series on fundamentals of generative AI and large language models. I am Vaishali. Today is my first session in this lecture series. I will start my session with conditional GAN. So, let us see what are the contents for today's session. The contents goes like quick review on GAN, motivation for conditional GAN, introduction to conditional GAN's architecture, training in GAN, objective function in conditional GAN, key difference between vanilla GAN and conditional GAN, and the applications.

[00:44] So, we will start with a quick review on the working of GAN. So, what exactly GANs are? So, GANs, as we all know, that GANs are generative adversarial network, which is a deep learning framework. And it consists of two main components, the generator and the discriminator, and they are trained in an adversarial manner. Now, when I tell they are trained in an adversarial manner, I mean that there are two entities who are there. In this case, it is the generator and the discriminator.

[01:24] And they are in opposition, and they are competing with each other. So, it's like a game. So, for example, if we take game of cricket, we have the bowler and the batsman. Okay? So, the batsman will always try to increase the number of runs that

he is doing, whereas the bowler will try to stop that one. So, the success of the bowler will be to reduce the number of runs that has been done by the batsman, where the while the batsman's success will be to increase the number of runs, okay? So, this is the adversarial framework and our GANs, they work in this adversarial framework and the components or the competitors in this uh GANs, they are the generator and the discriminator.

[02:14] So, GANs are trained through a minimax game where the generator and discriminator competes to improve each other's perfor- improve their performances, okay? So, we'll have a quick look into the uh architecture of GANs. So, it has the two components, discriminator and generator. These are basically the generator and the discriminator, they are basically uh neural networks, okay? And simple neural networks, they are. And we will show here how the training of the original GAN was done in using MNIST data set. So, here what happens is we have the digit five, okay? Which is taken from MNIST data set which have all the handwritten digits uh from zero to nine and in this case we have taken five as a digit, handwritten digit's image

[03:11] and it goes to the discriminator, okay? The discriminator gets trained on how a five digit looks like, okay? The generator the input to the generator in this case, in GANs, is a random noise, okay? We will come into what exactly is a random noise. So, right now we'll just keep it as it is an input to the generator, which is a random noise. And from this noise, the generator generates some image, okay? So, the generated image in this case, it has generated image of the digit seven.

[03:52] Now, this generated image, which we call as a fake image also, this goes to the discriminator. Now, the discriminator have both the original image, which is the digit five, and the image that is created by the generator, which is digit seven, okay? And then, the discriminator's task is to distinguish between the original image and the fake image. So, the discriminator tries to segregate these or classify these fake images as real or fake.

[04:34] Now, the discriminator may or may not be able to classify it properly in the first instance. So, we have the losses that are associated. So, we have the discriminator loss, which is the loss that is coming if the discriminator is not able to classify the fake image or distinguish the fake image from the real image, okay? So, from there, we get the discriminator loss. And this discriminator loss gets back propagated, so the back propagation happens here, and the discriminator starts improving itself to make it more accurate in classifying between real and fake image, okay?

[05:22] Whereas, there is another term called as the generator loss. What is the function of the generator here? The function of the generator is to generate images which looks very similar to the real image. Now, till the time the discriminator tells that the generated image or the fake image is not looking same as the original image, till that time the generator has to improve its efficiency or improve its performance.

So, the generator loss we get it here.

[05:55] If the discriminator is telling that the generated image is not as same as the real image. So, we get another loss, the generator generator loss, which gets back propagated, and the generator gets trained to improve its performance in each and iteration. Okay? So, this is the overall framework of generative adversarial network, which are the GANs. Now, I was telling that the generator takes the input as a random noise. Now, now what is random noise?

[06:36] Random noise is basically a vector in the latent space. Now, when I tell latent space, what exactly I mean by that? So, latent space it refers to an abstract and low-dimensional representation of data. Even though it is a low-dimensional representation, it captures all the essential features and the underlying pattern in the data. So, the latent space, think of a story book, which is a very long story book, and you have the summary of that one. So, you can think of the summary as a latent a space representation of that book. So, it is having concise uh you know, form.

[07:27] It is not as elaborated as the story book, but at the same time, it has all the important informations from the story. Okay? So, random noise is basically a uh vector taken from the latent space. Okay? And it's a latent vector. Uh it's a latent vector Z . Okay? Now, it looks like something like this. If it is 100 dimension, then it has 100 numbers. Okay? So, now this Z or the you know, the components of Z Z means the latent space. Capital Z is the latent space.

[08:15] And each component of the latent space is drawn independently from a normal distribution that is having mean zero and variance one. So, that is the criteria of a latent a vector. Okay? Now, the generator takes the input as the random noise, which is a latent which is a sample of latent vector from the latent space. And it gives a structured output. So, in our previous picture, we have seen the structured output is the digit seven.

[08:52] Okay? Now, a question comes. Is there any control on what gets generated by the generator? There is no control on the image that is generated by the generator. Now, you may ask me why? What is the reason for that one? So, we look into it. Latent space does not have an explicit semantic meaning in standard GANs. What do I mean by that one? It means, let us take a small example. Then it will that uh you know, textbook kind of a definition becomes much more clear when I explain it with a example. So, suppose you have a GAN which is trained with MNIST data set, which have zero to nine uh digits, handwritten digits in the data set.

[09:43] Now, we sample, suppose you take Z_1 , which is uh latent vector from the latent space space Z . So, and you pass it through the generator G , and you get an output three. Similarly, if you sample Z_2 , it passes through the generator G , and then it gives you sample it gives you the digit seven. Similarly, Z_3 gives you one. Okay? Now, can I generate a five digit five on demand? Do I know which Z to be taken so that

I get the five in the output from the generator? The straight answer is no, because there is no such explicit uh a mapping where we can say that this portion or this Z will give me five.

[10:44] So, there is absolutely no way where we can take a specific random noise and pass it through the generator to give me a output of my choice. Okay? So, we don't know which Z will give me five. Okay? And that comes as the main motivation for conditional GANs. Okay? So, what is conditional GAN? When we talk about conditional GAN, which is the topic of discussion in today's session, so, we know that in standard GANs, we cannot, you know, do any sort of conditioning. That is, the generator I cannot tell what noise to be given to generate a digit of my choice. So, there is no way we can do that one. So, is there a process by which we can condition the input to the generator so that the output is conditioned based on the

[11:46] input. Okay. So, this is once again I have kept the image of the standard GANs and just want before I go forward just want to emphasize this point that the generator is trained to maximize the probability of the discriminator to make a mistake. Okay, that is in standard GANs we do and conditional GAN also same thing. Discriminator is trained to maximize the probability of correct identification of the real and the fake images. Okay. So, this is the way the uh you know, generator and the discriminator are trained in GANs.

[12:25] And as I was telling before once again I'm reinforcing that same thing. The generator produce digits but without control over the digit it is producing. And if we use conditional GANs that enable a controlled and guided data generation. So, I can tell if I want five here as the uh you know, output from the generator what I should put as the input in the conditional GAN. I will put some conditioning parameter in the input of the generator so that the generator gives me exactly what I want. Okay. So, that was the motivation for conditional GANs.

[13:09] So, we will look into what is conditional GAN and how it works. What are the what is the architecture of conditional GAN. So, as I was telling that conditional GANs are basically conditioning your GANs with some extra information. Okay. Now, that conditioning your GANs with some additional information, this uh was first invented in this paper, Conditional Generative Adversarial Nets. This is the original paper of conditional GANs. And I urge all the participants to look into the paper, go through the paper. Now, when I take tell conditioning your GAN with some extra information, I mean the conditioning parameter will go to the generator as well as the discriminator, okay? So, both the generator as well as the

[14:03] discriminator will get the conditioning parameter. Now, in this case, suppose I am taking the MNIST data set, okay? So, the condition conditioning is done on the class label. So, suppose I am putting the digit seven. So, with the in the label, I will put it is I will tag it as seven, okay? So, that is the conditioning thing that happens

in conditional GAN. So, zero to nine digits from MNIST data set. And when we give a class label label, we are telling which digit or which image belongs to which digit, okay? Now, this conditioning parameter or the conditioning information » [snorts] » is given to the generator and discriminator while they are getting trained.

[15:00] So, this we call the call it as class labels, which is the conditioning parameter here, okay? So, along with the images, we are giving a class label or a tag kind of thing. And as I was telling, so the class label will come from from the MNIST data set. So, if the image is of the digit zero, I will tag it as zero. Okay? So, that information will be fed to the generator as well as the discriminator during the training process.

[15:40] Now, here in this case, it is you know, class label that we are giving, but GANs can also be conditioned on other modalities also like text prompts, uh images, you know, when in in the case of image-to-image translation. Okay? Now, we will have a look into how the conditional GAN works. So, in this case also, we have the generator and the discriminator. And we have the real images here, which is seven here. It goes to the discriminator.

[16:17] And we have the class label. So, the real images I'm telling as X. The class labels I'm telling as Y. Okay? And these class labels also goes to the discriminator. So, the discriminator have the real images X and the class label Y. Now, we have the noise sample from the latent space. So, noise is Z. This noise plus the class label, both goes to the generator. So, the input to the generator is noise plus class label. The input to the discriminator is the real image plus class label. Okay?

[17:00] Now, once I am giving the class label with the noise, we can see the generator is generating the correct digit, which is seven, because in the class label I have specified that I want an image of the digit seven. Okay? So, that is the conditioning that I have done for the generator. Now, these are the generated fake images. Now, if we see clearly or you know, if we see you know, carefully, we see this seven's structure and this seven structure, even though they are digital the digits are same, but the structures are different. So, what does the discriminator what does the discriminator do or what it is supposed to do? It has to see both the real seven and the generated seven and then tell whether the generated

[17:56] seven looks exactly similar to the real seven or not. Okay? So, what we do for that one? We give these generated sevens to the discriminator. So, now the discriminator is having the generated images or the fake images and the real images, both are going to the discriminator. And usually we give it in 50/50 50:50 ratio. Okay? Now, it goes and the discrimi- once the discriminator gets the real images as well as the fake images, it classifies them as real or fake. And once again, we have the same thing as we have seen last time. We have the discriminator loss and the generator loss that is coming out. And if the discriminator has made a mistake, then it this is the you know, back propagation the errors get back

[18:54] propagated through this path and the discriminator improves or gets trained iteratively in more iterations so that it can improve its performance. And if the generator is unable to fool the discriminator that if the generator's fake image is the discriminator is not telling as a, you know, real image or the generator is unable to fool the discriminator, we have the generator loss, which gets back propagated in this path and the generator parameters gets, you know, updated, weights gets updated and it keeps on uh training uh more and more till the performance improves, okay?

[19:40] Now, we will see step-by-step what happens in the training loop in GANs, okay? So, in the first step, it is same as what we have seen in the pictorial representation, but in step-by-step I will give you it's kind of a uh pipeline or a flow so that it becomes clear to all of you. So, in step one, we pick a real image X , which is seven in this case. This goes to the discriminator. Then pick its condition Y . And what is that one? In this case, it is label equal to seven.

[20:16] This also is given to the discriminator. Then in step two, so okay. So, the discriminator gets both the real image and the condition. Now, this $D(X, Y)$ it gives the probability score, which is given by the discriminator for this set. D , the real image and the class label, okay? Now, in step two, what do we do? I will come and explain this $D(X, Y)$ more in details in my subsequent slides. For right now, we will just understand this D is a probability score, which is given by the discriminator. It ranges between zero and one.

[21:10] Now, in step two, we have a random noise. Sample a random noise, and this random noise is shown here, and then it goes to the generator. And we provide the same condition Y to the generator also. So, the generator gets the noise and the condition. And we get $G(Z, Y)$, which is which means G stands for the output that is generated by the generator using Z , which is the noise, and the condition Y . Okay? So, the generator is generating an image from noise with the condition Y .

[21:53] Okay? So, it is given by $G(Z, Y)$. So, this basically tells generator output conditioned on Y . Then this generator output, we name them as the fake image, which is X_{fake} . Okay? So, this is we can just tell it is equivalent to your X_{fake} image. Okay? Now, this goes to the discriminator also. This goes to the discriminator. Okay? And the discriminator now have the real image, the generated image from uh the generator, and what is supposed to do, the discriminator sees and gives a score for the X_{fake} conditioned on Y .

[22:43] So, it gives a probability score. What is that probability score about? Whether it is looking similar to the image that is originally given to the discriminator, that is X , and whether it is conditioned properly on Y . Okay? So, these are the two things which the discriminator sees. Now, after this step, you know, while we are training the discriminator, now I'm coming, you know, into the probability scores which are given by the discriminator. Now, we will see this $D(X, Y)$ tends

to one for a real pair.

[23:23] That means, when the discriminator gets a real image and the conditioning label, the probability score should tend to one. Whereas, if you see the next one, $D(G(Z, Y))$, what does it mean? This is the output G within bracket Z {comma} Y , it gives the output of the generator which has taken as input the noise Z and the class label Y . So, this is the generator output and this is the class label. And then, for these two things, the discriminator is giving a probability score which tends to zero because it is a generated image. It is not the real image. So, it's a fake image, okay?

[24:21] So, the discriminator learns two things. What are the things that discriminator learns? Does the image match the condition? So, if the condition is seven, does the image match that seven condition? Does the image, whatever has come out of the generator, does it resemble the digit seven? And second is, is it real? So, it will see the fake image or the generated image and it will try to see is it similar or not similar? Which is just Does it exactly look same as the original image? So, these are the two things that the discriminator learns in the process of training.

[25:04] In step four, while the generator is getting trained, what happens? The generator The generator's output $G(Z, Y)$ conditioned on Y For this one, when the discriminator gives the probability score, the generator what it tries is it tries to fool the discriminator. That means it tries to make the discriminator think that this is a real image. This is not a fake image. It's a real image. And for that one, it tries to push this D or the probability score given by the discriminator for the image generated by the generator conditioned on Y to be towards one. Okay. So, the main uh thing that the generator we all know that the generator tries to do is to fool the discriminator. So, that is the reason it tries to push the probability

[26:07] score given by the discriminator towards one for the generated image which is conditioned on Y . So, the generator learns two things. Given label Y , generate an image that looks real and it matches the condition Y . So, these are the things that happen during the training of the discriminator and the generator. Then, in in step five, we can say the entire process gets repeated. The generator improves image quality, the generated image's quality, discriminator improves its detection efficiencies. It keeps on, you know, improving its detection efficiency so that it can detect or segregate correctly between a fake image and a real image. Eventually, the generator learns to produce realistic and label

[27:05] consistent samples. Okay? So, this is the overall training loop in conditional GANs. And then we will see the losses that are coming from the conditional GAN, the discriminator loss and the generator loss. So, we'll first look the discriminator loss. So, as I was telling, the goal of the discriminator is to predict real pairs as real and predict fake pairs as fake. This is clear. Now, this discriminator have two losses.

[27:38] The first one is the real loss. We will see what is this real loss. So, the real

loss is given mathematically. This is L for loss. D stands for the discriminator and this is the real loss that is coming from the discriminator. And this is given as the expectation value of $\log DXY$. And this X is taken from a distribution of the true data. So, I will explain each and every term here. So, this real loss measures how well the discriminator correctly classifies real data as real. Okay?

[28:21] And why this expectation value is given? Because it takes the expectation or average over the real data set. Okay? This P data of X , it represents the true data distribution. That is whatever the real images you have put, that data distribution is represented by P data X . And this $D X \{ \text{comma} \} Y$ basically gives the probability score of this pair being real given the image is X and the condition is Y . If you remember the image X is the original the real image, okay?

[29:06] So, this $D X \{ \text{comma} \} Y$ gives a probability score as I was telling which can, you know, vary between 0 and 1. Now, the discriminator wants $D X \{ \text{comma} \} Y$ to be 1. Why? Because X is the real image, Y is the condition. So, for a real image, the probability of the discriminator or the probability score given by the discriminator should be tending to 1. Okay. And if we see the negative sign is given for minimizing the loss.

[29:45] What So, if we see here for various values, just you can take the value of uh say the probability score as .9 and .2 and you will see what happens is if this probability value, suppose I'll just do it here. Suppose our uh this part $D X$ of Y equal to .9 then what happens? Then $\log$ of $D X \{ \text{comma} \} Y$, you can check this one. It is equal to -0.045, okay? And suppose you have $D X \{ \text{comma} \} Y$ equal to .2 then $\log$ of $D X \{ \text{comma} \} Y$ is equal to minus 0.69.

[30:48] Okay, 698 something. Now, if you see, discriminator wants $D X$ of Y tending to zero. So, this is what the discriminator wants. Now, when we do the $\log$ of that one, we get 0.045. And for 0.2, we get 0.698. That means what is happening here is the loss is more in this case compared to this one. So, but in uh you know, in deep learning, we always want to reduce to minimize the loss. So, we take a negative value here. Okay? So, the negative signs is given for minimizing the loss. Okay?

[31:36] Now, this is the real loss. There is a fake loss also associated with the real with a discriminator. So, discriminator has two losses. One is the real loss, and one is the fake loss. Real loss we have seen. We will now see what is this fake loss. Now, the fake loss's uh expression looks like $\exp$ expectation value of $\log 1 - D G Z Y Y$. So, this $G Z Y$ is the generated image by the generator with an input Z , which is nothing but your noise, and the class label Y .

[32:17] And the discriminator is giving a probability score for this image and the class label Y . Okay? Now, we will see the fake loss measures how well discriminator can detect generated data as fake. Okay? Now, this expectation value averages over all noise samples. So, the sample here belongs to PZZ , that is the noise samples, okay?

Because generator have generated the images from noise. Now, G of ZY I told produce the It is the produced uh image from the generator, and D G G ZY , {comma} Y gives the probability of being real given the sample is generated, and Y is the condition, okay?

[33:15] Now, discriminator wants Discriminator will never want the probability of this one to be real because discriminator knows that it is a generated sample. So, the discriminator would want D of GZY , Y should tend to zero. Okay? So, this is what the discriminator wants. So, this loss comes from the fake images. And the total discriminator loss is given by the sum of the fake loss and the real loss. Okay? So, this is the real loss, and this is the fake loss, okay?

[33:57] Now, we will see the generator loss in cGAN. What is this generator loss? So, we all know the generator tries to fool the discriminator by producing samples which look very realistic for that specific condition. We know that one. Now, there is a standard generator loss which is given by L_G is represents the generator loss expectation value over Z and $Y \log 1 - D(GZ|Y)$. Once again, this is the generated image conditioned on Y . This D gives the discriminator's score. Discriminator gives a probability score that is given by your D of this entire thing. And one minus this one.

[34:45] So, this is the standard generator loss. But, if we use this one, there is a vanishing gradient problem which occurs. And that is the reason in practical case cases, we don't take into take this loss, we take the loss as L_G equal to expectation value over $Z \log D(GZ|Y)$. Okay. So, that is the expression. We are not going to the derivation, but this is the expression for the uh generator loss. And this L_G measures how well the generator can fool the discriminator.

[35:26] This expectation value is over all the noise samples that negative term as as uh I have told before also, it is added to minimize the loss. And the probability score that the discriminator assigns to the the uh generator's fake image, the generator wants it to be tending to one. Okay. So, this is the probability score and the generator wants So, this generator wants generator wants this probability score should go to one.

[36:01] Whereas, the discriminator will try to make the probability score towards zero because this is a generated image. Okay. So, as I have to wrote also, generator wants to maximize this value. Okay. Now, then we have seen the discriminator loss, the generator loss, then we are coming to the objective function of C GAN. So, in C-GAN, the overall objective is a minimax game. It's a minimax game between generator D uh G and the discriminator D , both conditioned on Y , okay? So, we will see This is the objective function. It looks like this. I'll uh explain, you know, um each and every term of this expression.

[36:49] So, $\min_G \max_D$ Sorry, $\min_G \max_D$. The discriminator tries to maximize its ability to distinguish between the real and the fake, while the generator tries to

minimize the discriminator's success, okay? So, that is captured by this minmax objective function. And the discriminator, if you see $\max D$, the discriminator actually tries to maximize the objective function, while the the generator tries to minimize the objective function, okay?

[37:24] So, V is the overall objective function that both the networks are competing to optimize. So, V is the objective fun function. Okay, as I I told um reinforcing that one, the discriminator tries to maximize the objective function, while the generator tries to minimize it, okay? And this term, we know D_{XY} , that is the discriminator's probability score generated by the discriminator for a real image conditioned on Y , encourages the discriminator to classify real samples correctly.

[37:59] And this term this is the generated image by the generator. Okay, which has taken input Z and Y , and this term pushes the generator to produce samples that the class that the discriminator can classify as real, okay? So, this is the overall objective function in conditional GAN. As the training progresses, both generator and discriminator improves. And this competition or the adversarial framework produces results. And the generator Uh what are the results that are produced in this adversarial framework? That the generator produces more realistic data.

[38:45] And these realistic data are conditioned for that specific condition. Okay? So, they look very realistic at the same time they follow the condition that has been given. Okay? So, this is about the objective function of CGANs. A quick look into the key differences between vanilla GAN and conditional GAN. So, vanilla GAN we know that is the uh normal GAN. Generator's output depend only on the random noise, whereas in conditional GANs, it depends both on the noise as well as the um class label or the conditioning parameter.

[39:21] Here, D_X gives the probability that the X input X is real from data. Whereas, here the probability score which is generated by the um discriminator, it uh takes both the input real input X as well as the condition. So, it sees that the probability that the input X is real, that is it is from the actual data set, and it matches with the condition Y . Okay? So, both the things it sees. And in vanilla GAN, we see there is no control on generation. And here, it is a control generation.

[40:01] And applications of conditional GANs. So, we have the the the entire thing that I have told, it is based on the MNIST data set. Now, there are other applications, image to image translation. So, you have a sketch and you take you want a photograph from that sketch that can be done with conditional GANs. Then from daytime image to nighttime image, that is possible. Text to image translation like create a red bird with blue wings. So, it has created something. So, text to image translation is possible. Then generating high resolution images for from low resolution input. So, this is a low resolution input. You can see much more details in this image. So, from low to high resolution images and converting photo to painting

[40:53] of different styles. In this case, it is Van Gogh style. So, there is a painting here. It got converted to this one. A particular style of painting, okay? And so, these are the various applications of conditional GAN. So, with this one, I come to the end of the session today. So, before I conclude, let me summarize or just go through once what all we have learned today. So, we have taken an overview or a quick recap of the normal GANs, how it works, what is the adversarial framework, how it, you know, helps the generator and the discriminator to compete with each other and improve their performances. And then we have looked into what is the limitation of standard GANs. Uh there we don't have

[41:40] any control on the output that is generated by the generator. So, for that one, that was the main motivation for conditional GANs. Now, in conditional GANs, what we have done, we have given whatever we were we were giving to the standard GAN, with that one, we are giving a class label or a conditioning parameter or conditioning input parameter input that goes to both the generator and the discriminator. So, the whatever the generator generates, it is conditioned on the input. It conditioned on a specific uh you know whatever the user wants.

[42:17] And that is the way the output of the generator becomes controlled. Okay, with the given condition. Then we have seen what is the architectural you know difference between normal standard GAN and the conditional GAN. We have looked into the losses that are there associated with the generator and the discriminator. We have looked into the entire pipeline how the images are taken by the discriminator with the class label. The generator takes the noise and the class label. Generator generates the fake image. That also goes to the discriminator. The discriminator have both the real image and the fake image and the class label. And from there it segregates between real and fake images. And then to reduce the loss the losses of the

[43:09] discriminator gets back propagated to the discriminator and the discriminator keeps improving. The generator loss gets back propagated through the network and the generator keeps improving over iterations. Okay, so this is the training pipeline that we have seen and we have seen what are the mathematical forms of the different kind of losses that are coming in the generator and the discriminator. And finally we have seen how the objective function which is a minimax objective function which the discriminator wants to maximize the objective function and the generator wants to minimize that one. So we have looked into and then finally we have looked into the applications, the various applications of conditional GAN.

[43:54] Conditioning can happen with the um class labels. It can happen with images. It can happen with text. It can happen with a various other modalities. So we have we I just had a look into that one. And with this one, I conclude today's session. In the next session, we will talk about Pix2Pix GAN, which is specifically designed for image-to-image translation. As I was showing in one of my applications, that from

sketch to an a photo, how the conditional GAN is changing it.

[44:28] So, Pix2Pix GAN is a specialized architecture of conditional GAN, which is used for image-to-image translation. That will be the topic of my next discussion. Till that time, happy learning. Thank you all.